

1. Which conclusion is correct about a series whose n -th partial sum is

(4 Points)

$$s_n = \frac{5 - 8n}{2 + 9n}$$

- ☒ It is convergent with sum $-\frac{8}{9}$ ✓
- ☐ It is convergent with sum $\frac{5}{2}$
- ☐ It is convergent with sum 1
- ☐ It is divergent

2. Which of the following is correct for the geometric series

(5 Points)

$$-\frac{1}{5} + \frac{1}{25} - \frac{1}{125} + \frac{1}{625} - \dots$$

- ☒ It is convergent with sum $-\frac{1}{6}$ ✓
- ☐ It is convergent with sum $-\frac{1}{5}$
- ☐ It is convergent with sum $-\frac{1}{4}$
- ☐ It is divergent

3. Which of the following is true for the series

(5 Points)

$$\sum_{n=0}^{\infty} 7 \cdot \frac{2^{n-1}}{5^n}$$

- ☒ It is convergent with sum $\frac{35}{6}$ ✓
- ☐ It is convergent with sum $\frac{7}{5}$
- ☐ It is convergent with sum $\frac{2}{5}$
- ☐ It is divergent

4. Which of the following arguments is correct for the series

(4 Points)

$$\sum_{n=1}^{\infty} \cos\left(\frac{5}{n}\right)$$

- ☒ It is divergent since $\lim_{n \rightarrow \infty} \cos\left(\frac{5}{n}\right) \neq 0$. ✓
- ☐ It is divergent since $\sum_{n=1}^{\infty} \frac{5}{n}$ is divergent and $\cos$ is a continuous function.
- ☐ It is convergent by the Alternating Series Test.
- ☐ It is convergent by the Integral Test.

5. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$$

- ☒ The Integral Test applies and concludes that the series is divergent. ✓
- ☐ The Integral Test applies and concludes that the series is convergent.
- ☐ $f(x) = \frac{x}{x^2 + 1}$ is not decreasing on $[1, \infty)$, so The Integral Test does not apply.
- ☐ $f(x) = \frac{x}{x^2 + 1}$ is not positive on $[1, \infty)$, so The Integral Test does not apply.

6. Determine which one of the series below is divergent.

(4 Points)

- ☒ $\sum_{n=1}^{\infty} \frac{1}{n^{0.3}}$ ✓
- ☐ $\sum_{n=1}^{\infty} n^{-5\pi}$
- ☐ $\sum_{n=1}^{\infty} \frac{1}{n^e}$
- ☐ $\sum_{n=1}^{\infty} \frac{1}{n^3}$

7. Find all positive values of c for which the following series converges.

(5 Points)

$$\sum_{n=1}^{\infty} c^{\ln n}$$

- ☒ $0 < c < \frac{1}{e}$ ✓
- ☐ $c = e$
- ☐ $0 < c < e$
- ☐ $c > \frac{1}{e}$

8. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^2 + 1}$$

- ☒ It is convergent by the Comparison Test with the p -series with $p = 2$. ✓
- ☐ It is convergent by the Comparison Test with the p -series with $p = \frac{1}{2}$.
- ☐ It is divergent by the Comparison Test with the p -series with $p = 1$.
- ☐ It is divergent by the Test for Divergence.

9. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} \frac{e^{\frac{1}{n}}}{n^{1+\frac{1}{2}}}$$

- ☒ It is divergent using the Limit Comparison Test with the series $\sum_{n=1}^{\infty} \frac{1}{n}$. ✓
- ☐ It is convergent using the (Basic) Comparison Test with the series $\sum_{n=1}^{\infty} \frac{e}{n^2}$.
- ☐ It is divergent by the Test for Divergence.
- ☐ It is convergent using the Limit Comparison Test with the series $\sum_{n=1}^{\infty} \frac{1}{n^{1.1}}$.

10. Which of the following statements is false?

(5 Points)

- ☒ If $a_n > 0$ and $\lim_{n \rightarrow \infty} n^2 a_n = 2$, then $\sum_{n=1}^{\infty} a_n$ is divergent. ✓
- ☐ If $a_n > 0$ and $\lim_{n \rightarrow \infty} n a_n = 2$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- ☐ If $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty} \sin(a_n)$ is convergent.
- ☐ If $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty} \cos(a_n)$ is divergent.

11. Find all values of p for which the following series converges.

(4 Points)

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{(\ln k)^p}$$

- ☒ $p > 0$ ✓
- ☐ $p < 0$
- ☐ $p > -1$
- ☐ $p = -1$

12. Which of the following is correct about the statement

(4 Points)

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

- ☒ It is always true. ✓
- ☐ It is always false.
- ☐ It is sometimes true.
- ☐ It depends on a_n .

13. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} \frac{n^n}{10^{3n+1}}$$

- ☒ It is divergent by the Root Test. ✓
- ☐ It is divergent by comparison with a p -series.
- ☐ It is convergent by the Ratio Test.
- ☐ It is convergent by comparison with a geometric series.

14. Which one of the following series is absolutely convergent?

(5 Points)

- ☒ $\sum_{k=1}^{\infty} \frac{(-1)^k \arctan k}{k^4}$ ✓
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{\sqrt[k]{k}}$
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$
- ☐ $\sum_{k=1}^{\infty} \frac{\cos k\pi}{k^{-3}}$

15. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} (-1)^n \tan\left(\frac{1}{n}\right)$$

- ☒ It is conditionally convergent. ✓
- ☐ It is absolutely convergent.
- ☐ It is divergent.
- ☐ None of these.

16. Find the radius of convergence of the power series.

(5 Points)

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x)^{2n}}{(2n)!}$$

- ☒ ∞ ✓
- ☐ 2
- ☐ $\frac{1}{2}$
- ☐ 1

17. Find the interval of convergence of the power series

(5 Points)

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x-8)^n}{n}$$

- ☒ $(7, 9]$ ✓
- ☐ $[7, 9)$
- ☐ $(-8, 8]$
- ☐ $[-8, 8]$

18. Use the power series representation centered at 0 of $f(x)$ to find the sum s .

(5 Points)

$$f(x) = \frac{1}{(1-x)^2}, \quad s = \sum_{n=1}^{\infty} \frac{n}{3^{n-1}}.$$

- ☒ $s = \frac{9}{4}$ ✓
- ☐ $s = \frac{1}{4}$
- ☐ $s = \frac{3}{2}$

19. Use the power series representation centered at 0 of $\arctan(x)$ to find the sum of the series

(5 Points)

$$\sum_{n=0}^{\infty} (-1)^n \frac{2^{2n+1}}{(2n+1)7^{2n}}$$

- ☒ $7 \arctan\left(\frac{2}{7}\right)$ ✓
- ☐ $\frac{7}{2} \arctan\left(\frac{7}{2}\right)$
- ☐ $2 \arctan\left(\frac{2}{7}\right)$
- ☐ $\arctan\left(\frac{7}{5}\right)$

20. Which of the following pairs is a power series representation and its radius of convergence for the function

(5 Points)

$$f(x) = \frac{x}{4x+1}$$

- ☒ $\sum_{k=0}^{\infty} (-1)^k 2^{2k} x^{k+1}, \quad R = \frac{1}{4}$ ✓
- ☐ $\sum_{k=0}^{\infty} (-1)^k 2^k x^{k+1}, \quad R = \frac{1}{2}$
- ☐ $\sum_{k=0}^{\infty} (-1)^k 2^{2k} x^{k+1}, \quad R = 4$
- ☐ $\sum_{k=0}^{\infty} (-1)^k 2^k x^{k+1}, \quad R = 2$

21. Evaluate the indefinite integral. What is radius of convergence?

(5 Points)

$$I = \int \left(\frac{x}{5} - \frac{x^2}{2.5^2} + \frac{x^3}{3.5^3} - \frac{x^4}{4.5^4} + \dots \right) dx, \quad \text{where radius of convergence } R > 0.$$

- ☒ $I = C + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{n+1}}{n(n+1)5^n}, \quad R = 5$ ✓
- ☐ $I = C + \sum_{n=1}^{\infty} \frac{x^{n+2}}{n(n+2)5^n}, \quad R = 1$
- ☐ $I = C + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n 5^n}, \quad R = \frac{1}{5}$